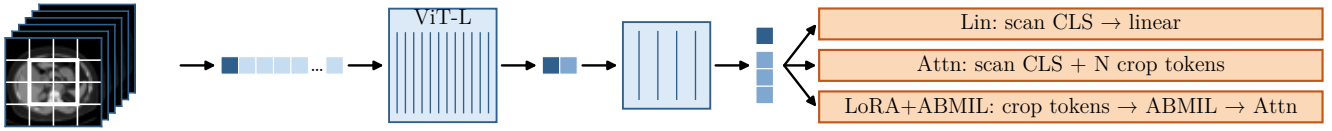
(a) SPECTRE: crops, per-crop ViT-L, feature combiner



$512^2 \times 256$, 0.5×0.5×1 mm, $4^3 = 64$ crops (train: $2^3 = 8$)

$128^2 \times 64$ crop: 16×16×8 patches, 512 + CLS

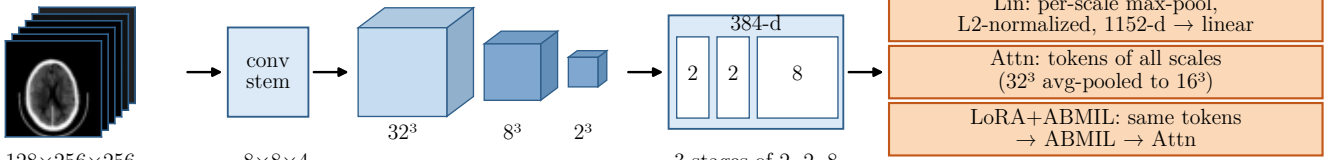
24 blocks, 1080-d, per crop

[CLS ‖ mean], 2160-d

combiner: 4 blocks over crops

scan CLS + N crop tokens

(b) Pillar-0 (head CT): multi-window input, conv stem, three-stage multi-scale encoder



32^3 8^3 2^3

128×256×256, 0.5×0.5×1.25 mm; 11 channels (10 HU windows, min–max)

8×8×4 patches, 32^3 grid

tokens at 3 scales (max-pool by 4 per axis), attention within and across

3 stages of 2, 2, 8 blocks; scales 32^3, 8^3, 2^3 leave after stages 1, 2, 3

(c) What each classification method trains

	patch emb.	early blocks W_{qkv}	early blocks W_{proj}	early blocks MLP, norms	late blocks W_{qkv}	late blocks W_{proj}	late blocks MLP, norms	head	steps	encoder parameters updated: TAP-CT	SPECTRE	Pillar-0
Lin								pooled → linear	8000	0	0	0
Attn								attentive probe	8000	0	0	0
LoRA								attentive probe	8000	0.59 M	2.4 M	0.55 M
LoRA$_{\text{late}}$								attentive probe	8000	0.20 M	1.0 M	0.25 M
LoRA+ABMIL								ABMIL → Attn	8000	0.59 M	2.4 M	0.55 M
DoRA *(+ m)*		m			m			attentive probe	8000	0.62 M	2.5 M	0.57 M
rsLoRA *($\alpha/\sqrt{r}$)*								attentive probe	8000	0.59 M	2.4 M	0.55 M
Lin→LoRA					2			linear	4000 + 4000	0.20 M	1.0 M	0.25 M
Attn→LoRA					2			attentive probe	4000 + 4000	0.20 M	1.0 M	0.25 M
Attn-only FT								attentive probe	8000	28 M	131 M	11 M
Full FT								attentive probe	8000	87 M	397 M	45 M

frozen rank-16 adapter (m: DoRA magnitude; 2: added in stage 2) trained in full